Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Forward Deployed Engineer with 8+ years shipping production code into enterprise integration layers and owning complex client onboardings end-to-end. At Entrata, I built and maintained the integration layer connecting property management operations across 5+ dependent enterprise systems -- REST APIs, webhook endpoints, event-driven async pipelines (RabbitMQ), CRM connectors, and scheduling workflows for 100K+ monthly active users. I take apart systems to see how they actually work: I've debugged production failures with no stack trace and returned with the root cause. Now building production agentic AI: Leasel is an 8-agent LangGraph pipeline with multi-system orchestration, custom business logic, and human-in-the-loop escalation (93.6% precision). Strong in Python, TypeScript, React, and production debugging across the full stack. AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Enterprise Integration Layer | API Integrations & Webhooks | CRM & Scheduling System Connectors | Event-Driven Async Pipelines | Agentic AI & Multi-Agent Orchestration | Client Onboarding & Technical Consulting | Production Debugging & Incident Response | Python, TypeScript & React

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building agentic AI pipelines and integrating data systems with back-end and client-facing APIs; debugging integration issues across the application stack; turning recurring implementation patterns into reusable product components.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Owned complex client onboardings end-to-end:** embedded with operations teams, translated multi-system business logic into a LangGraph agentic pipeline (parse → plan → retrieve → validate → routing), built direct API integrations with client enterprise systems, deployed to production, and served as primary technical consultant during rollout.
- Built event-driven validation layer enforcing business rules before output delivery; implemented RAG pipeline with cited compliance decisions; turned recurring implementation learnings into reusable product components handed off to the core engineering team.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; designed operational dashboards (+40% comprehension).

Entrata (Enterprise PropTech SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Built and owned the integration layer connecting property management operations across 5+ enterprise systems** -- REST API design and implementation, webhook endpoints, event-driven async pipelines (RabbitMQ) with idempotency and retry guarantees, CRM connectors, scheduling and eligibility workflows, and third-party data source integrations for 100K+ monthly active users at 99.5% uptime. Zero production incidents caused by integration failures across 7 years.
- **Owned technically complex onboardings end-to-end** -- served as primary technical consultant for new affordable housing customers; mapped their multi-system business logic to platform capabilities; wrote per-team technical specs; coordinated cross-system dependencies; validated integrations end-to-end before go-live. Zero compliance failures at go-live across every deployment.
- Debugged production failures across the full integration stack -- APIs, async pipelines, database queries, document processing -- typically with minimal context; built internal debugging tools that reduced mean debug time 40%; optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers via 300+ code reviews.

PROJECTS

Leasel: Multi-Agent AI Platform with Enterprise System Integration

LangGraph 8-agent -- FastAPI -- RAG -- Multi-System Orchestration -- HITL

Production agentic AI platform built end-to-end with no PM and no roadmap: 8-agent LangGraph pipeline with durable state management (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) orchestrating across structured databases, document stores, configuration systems, and regulatory rule engines. Multi-system orchestration with custom business logic at each node; human-in-the-loop escalation when confidence is too low; guardrails blocking unsupported outputs. Results: 93.6% precision / 87.9% F1 on 1,240 test cases; 100% citation correctness.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP Integration Layer for Clinical Records Systems

Flask -- LLMs -- BERT -- REST API -- Webhooks -- Docker

Built an integration layer translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings -- direct API integration with clinical records systems, webhook-driven event routing, HIPAA-compliant RBAC across multi-system data stores. Owned technically complex onboarding for clinical staff; drove adoption across workflows. Measurable outcome: ~60% reduction in manual retrieval time.

github.com/venkatalakshmiparimala/AI_on_FHIR

AgentProbe: Open-Source AI Agent Evaluation & Observability Toolkit

FastAPI -- LangChain -- ChromaDB -- CI/CD -- Production Tooling

Production evaluation toolkit for AI agents and integration pipelines: 15-endpoint FastAPI backend, 5-metric eval engine, CI/CD regression gate, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), observability dashboard. Built to measure whether deployed AI integrations are achieving their operational objectives across 1,240+ test cases. Turns implementation learnings into reusable tooling.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

Integration & APIs: REST APIs, Webhook Endpoints, Event-Driven Architecture, RabbitMQ, API Integrations, CRM Integrations, Scheduling System Connectors, Third-Party Integrations, Enterprise Integration Layer, Multi-System Orchestration, OAuth, SAML, OpenAPI, WebSockets, Async Pipelines, Idempotency & Retry Logic

AI & Agents: LangChain, LangGraph, Agentic AI, Voice AI, Multi-Agent Orchestration, LLMs, RAG Pipelines, Prompt Engineering, Function Calling, AI Guardrails, HITL, Observability, FastAPI, MCP, OpenAI API, Anthropic API

Languages & Frameworks: Python, TypeScript, JavaScript, React.js, Node.js, SQL, FastAPI, Flask, Django, Java, Spring Boot, PHP

Cloud & Infra: AWS (S3, EC2, Lambda, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, Redis, Microservices, Linux, Production Debugging, Distributed Systems